

Overview of Fargues - Scholze

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Construction of the spectral action

$F | \mathbb{A}_F$ fin. ext., G/F , Λ a $\mathbb{Z}_\ell$ -algebra

Goal. Construct an action of $\text{Exc}(G, \Lambda) = \varinjlim_{F_n \rightarrow W_F} \mathcal{O}(\mathbb{Z}^1(F_n, \hat{G}))$
on the category of smooth Λ -reps of $G(F)$.
identity functor of

$$\text{Exc}(G, \Lambda) \rightarrow \mathcal{Z}(G(F), \Lambda)$$

FS: Instead construct an action on $\text{id}(D(\text{Bun}_G, \Lambda))$.

Fact. - $\text{Bun}_G \xleftarrow[\text{open}]{\circ/G(F)}$ inducing fully faithful functor

$$D_{\text{sm}}(G(F), \Lambda) \hookrightarrow D(\text{Bun}_G, \Lambda).$$

Essentially v. Lafforgue's idea: $\begin{array}{c} X \\ | \\ \mathbb{F}_q \end{array}$, $\begin{array}{c} G \\ \searrow \\ X \end{array}$ unramified

$$\begin{array}{c} G \\ | \\ F \end{array} \longrightarrow \begin{array}{c} X_{FF} \\ | \\ \text{Spa } F \end{array}$$

We work over the cat. of perfectoid spaces $/\overline{\mathbb{F}_p}$.

$$\begin{aligned} \text{Spa}(R, R^+) &\mapsto X_{FF}(S) = \left(\text{Spa } W_{\overline{\mathbb{F}_p}}(R^+) - V(P \cdot [W]) \right) / \phi_q^\mathbb{Z} \\ \text{Bun}_G(S) &= \{ G\text{-torsors on } X_{FF}(S) \} \end{aligned}$$

$$\text{Div}^d(S) = \{ \text{degree } d \text{ divisors on } X_{FF}(S) \}$$

$$\text{Spa } \overline{\mathbb{F}_p} \hookrightarrow \text{Div}^1 = \left(\text{Spa } F / \phi_q^\mathbb{Z} \right) \times_{\text{Spa } \overline{\mathbb{F}_p}} \text{Spa } \overline{\mathbb{F}_p} = \left[\text{Spa } C / \underline{W_F} \right] \begin{pmatrix} \text{twisted } W_F\text{-action} \\ C = \mathbb{C}_p = \hat{\mathbb{C}_p} \end{pmatrix}$$

(assume Λ is l -power torsion) $\forall X$ a v -stack

$$\exists D_{\text{ét}}(X, \Lambda) \xrightarrow{\text{full subcat.}} D(X_v, \Lambda)$$

$*$ - pullback / pushforward, $\otimes$, $R\text{Hom}$, $!$ - pullback / pushforward

for " l -fine" maps

$$D_{\text{ét}}(\text{Spd } \overline{\mathbb{F}}_p, \Lambda) \simeq D(\Lambda) \simeq D_{\text{ét}}(\text{Spd } \mathbb{C}, \Lambda)$$

Th. $D_{\text{ét}}(\text{Bun}_G, \Lambda)$ has a semiorthogonal decomposition into $D_{\text{ét}}(\text{Bun}_G^{[b]}, \Lambda)$
for $b \in B(\mathbb{Q}_p, G)$.

$\xrightarrow{\Sigma}$

§ Geometric Satake (S affinoid)

given a (deg d) divisor $D \subset X_{\text{FF}, S}$, form $B^+ = \varprojlim_n H^0(\mathcal{O}_{X_{\text{FF}, S}} / I_D^n)$
 $B = B^+[\frac{1}{I_D}]$

Def. Local Hecke stack $\text{Hck}_G^d \rightarrow \text{Div}^d$

given D , parametrizing $(P, Q, P \otimes B \simeq Q \otimes B)$

P, Q are G -torsors on B^+

fibres look like $L^+G \backslash L_G / L^+G$ (usual Hecke stack)

given $S \rightarrow \text{Div}^d$, write $\text{Hck}_{G, S}^d := \text{Hck}_G^d \times_{\text{Div}^d} S$

Def. Inside $D_{\text{ét}}(\text{Hck}_{G, S}^d, \Lambda)$

locally on S $D_{\text{ét}}(\text{Hck}_{G, S}^d, \Lambda)^{\text{bd}}$

has qc-supp

$D_{\text{ét}}^{\text{ULA}}(\text{Hck}_{G, S}^d, \Lambda)$

$\text{Sat}(\text{Hck}_{G, S}^d, \Lambda) \leftarrow$ flat perverse sheaves, i.e.
 $\forall \Lambda\text{-mod } M, A \bigoplus_{\Lambda} M$ is perverse.
ULA after pulling back to $\text{Hck}_{G, S}^d$.
 $\leftarrow$ perverse t-structure $A \in \text{PD}^{\leq 0} \Leftrightarrow \forall \mu_1, \dots, \mu_r$, restricting to corresponding stratum in $\deg \leq -\sum \langle 2p_i, \mu_i \rangle$

Restrict to $S = (\text{Div}^1)^I \rightarrow \text{Div}^{|I|}$

Write $\text{Hck}_{h,S}^{|I|} = \text{Hck}_h^I$.

Th. (Geometric Satake) If Λ is a torsion $\mathbb{Z}_\ell[\Gamma_a]$ -algebra, there is an equiv. of cats

$$\text{Sat}(\text{Hck}_h^I, \Lambda) \simeq \text{Rep}_{\Lambda}^{\text{fin. proj.}}((\hat{G} \times_{W_F})^I)$$

$$\textcircled{1} F_h^I = \bigoplus_{i \in \mathbb{Z}} H^i(\omega_h^I, -) : \text{Sat}(\text{Hck}_h^I, \Lambda) \rightarrow \text{Det}((\text{Div}^1)^I, \Lambda)$$

agrees with restriction along

$$\text{is } \leftarrow \text{Drinfeld Lemma} \\ D(W_F^I, \Lambda)$$

$$W_F^I \subset (\hat{G} \times_{W_F})^I$$

$$\textcircled{2} \text{ functorial w.r.t. } I : f: I \rightarrow J$$

$$\text{Hck}_h^I \times_{(\text{Div}^1)^I} (\text{Div}^1)^J \simeq \text{Hck}_h^J$$

$$\text{Sat}(\text{Hck}_h^I) \rightarrow \text{Sat}(\text{Hck}_h^J)$$

$$\textcircled{3} I = I_1 \sqcup I_2 \sqcup \dots \sqcup I_k,$$

$$\text{Hck}_h^I \supset_{\text{open}} \text{Hck}^I; I_1, \dots, I_k \subset \text{Hck}^{I_1} \times \dots \times \text{Hck}^{I_k} \quad \text{"fusion"}$$

$\textcircled{4}$ corresponding sym. monoidal str. on $\text{Sat}(\text{Hck}_h^I, \Lambda)$ agrees with the convolution structure.

Def. The global Hecke stack is the moduli $\text{Hck}_h^d = \{(P, \alpha, \rho |_{X_{FF-D}} \simeq \alpha |_{X_{FF-D}})^D\}$

$\swarrow \quad \searrow$
 $\text{Bun}_h \quad \text{Bun}_h \times \text{Div}^d$
 $\downarrow \quad \downarrow$
 $\text{local Hck}_h^d \quad (D, \rho|_{B^+}, \alpha|_{B^+}, z|_B)$
 $\rho, \alpha \quad G\text{-torsors on } X_{FF}.$

Given $V \in \text{Rep}_{\Lambda}^{\text{t.m.s.}}((\hat{G} \times W_F)^I)$

$\rightsquigarrow S_V \in D(\text{Hck}_{\hat{G}}^I, \Lambda)$ by geom. Satake

Def. The Hecke operator T_V is

$$T_V = p_{2!} \left(p_1^* (-) \otimes_{\Lambda} S_V \right)$$

$$: \text{Dét}(\text{Bun}_{\hat{G}}, \Lambda) \rightarrow \text{Dét}(\text{Bun}_{\hat{G}} \times (\text{Div}^1)^I, \Lambda)$$

$$\stackrel{\text{Drinfel'd}}{=} \underset{\text{Lemme}}{\text{Dét}}(\text{Bun}_{\hat{G}}, \Lambda)^{B W_F^I}$$

$(W_F^I\text{-equiv.})$

$$\begin{array}{ccc} & \text{Hck}_{\hat{G}}^I & \\ p_1 \swarrow & & \searrow p_2 \\ \text{Bun}_{\hat{G}} & & \text{Bun}_{\hat{G}} \times (\text{Div}^1)^I \end{array}$$

abuse notation: consider

$$T_V : \text{Dét}(\text{Bun}_{\hat{G}}, \Lambda) \longrightarrow \text{Dét}(\text{Bun}_{\hat{G}}, \Lambda)$$

by pulling back along

but remember the W_F^I -action.

$$\text{Spd } C \xrightarrow{\Delta} (\text{Div}^1)^I$$

Rem. Categorical local Langlands predicts

$$\text{Dét}(\text{Bun}_{\hat{G}}, \Lambda) \longleftrightarrow \text{coh}(\hat{\mathcal{X}}_{\hat{G}}) \quad \mathbb{Z}^1(W_F^0, \hat{G}) \text{ // } \hat{G}$$

$\hookrightarrow$

T_V

$\longleftrightarrow$

$\hookrightarrow$

$$(-) \otimes \mathcal{E}_V$$

$\mathcal{E}_V$ v.b. on $\hat{\mathcal{X}}_{\hat{G}}$ induced by

$$W_F \rightarrow \hat{G} \times W_F \xrightarrow{V_i} \text{GL}_n(\Lambda)$$

gives a W_F -action on each $\mathcal{E}_{V_i}$

$$V = \boxtimes V_i$$

T_V

$\longleftrightarrow$

$$(-) \otimes \mathcal{E}_{V_1} \otimes \dots \otimes \mathcal{E}_{V_d}$$

$\hookrightarrow$
 W_F

$\hookrightarrow$
 W_F

$$f \in \text{End}(\text{id}_{\text{D}(\text{Bun}_{\hat{G}})}) \longleftrightarrow$$

$$F \text{ coh. sheaf, } f \in H^0(\mathcal{O}_{\hat{\mathcal{X}}_{\hat{G}}}), F \xrightarrow{f} F$$

Pages

Given $V \in \text{Rep}_\Delta((\hat{G} \rtimes Q)^I)$ where $Q \leftarrow W_F$
 $\hat{G} \twoheadrightarrow$ finite quotient
 $\left. \begin{array}{c} \} \\ T_V \end{array} \right\}$

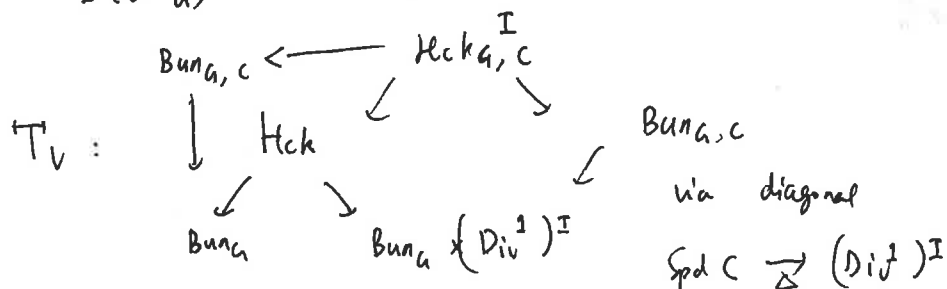
- $\alpha: 1 \rightarrow V$ as reps of $\hat{G}$, $\hat{G} \trianglelefteq (\hat{G} \rtimes Q)^I$

$\beta: V \rightarrow 1$ $\xrightarrow{\quad} \quad \xrightarrow{\quad}$

- $\gamma_i \in W_F^\circ \subset W_F$ for $i \in I$.

$$\text{id}_{D(\text{Bun}_G)} = T_1 \xrightarrow{T_\alpha} T_V \xrightarrow{(\gamma_i)} T_V \xrightarrow{T_\beta} T_1 = \text{id}_{D(\text{Bun}_G)}$$

"Excursion operators"



T_V only depends on $V|_{\hat{G}}$ ($\hat{G} \trianglelefteq (\hat{G} \rtimes Q)^I$)

λ only uses the "classical" geometric Satake.

$$(V, \alpha, \beta, (\gamma_i)) \rightsquigarrow \text{id}_{D(\text{Bun}_G)} \rightarrow \text{id}_{D(\text{Bun}_G)}$$

$$\searrow \quad \nearrow$$

$$f \in \mathcal{O}(\hat{G} \backslash (\hat{G} \rtimes Q)^I // \hat{G}), (\gamma_i)$$

given $\alpha, \beta, V, (\gamma_i)_{i \in I}$ in $\hat{G} \rtimes Q$.

$$\text{Consider } \left(\Lambda \xrightarrow{\alpha} V \xrightarrow{(\gamma_i)} V \xrightarrow{\beta} \Lambda \right) \in \Lambda$$

$\hat{G}$ -invariant under left Δ - $\hat{G}$ -action on (γ_i)

$$\rightsquigarrow \text{defines } f(V, \alpha, \beta) \in \mathcal{O}(\hat{G} \backslash (\hat{G} \rtimes Q)^I // \hat{G}).$$

- $(\alpha, \beta, \nu), (\alpha', \beta', \nu'), \quad s: V \rightarrow V', \quad (\hat{G} \rtimes Q)^I\text{-equivariant}$

$$\begin{array}{ccccccc} T_1 & \xrightarrow{\alpha} & T_\nu & \xrightarrow{(r_i)} & T_\nu & \xrightarrow{\beta} & T_1 \\ \parallel & & \downarrow s & & \downarrow s & & \parallel \\ T_1 & \xrightarrow{\alpha'} & T_{\nu'} & \xrightarrow{(r_i')} & T_{\nu'} & \xrightarrow{\beta'} & T_1 \end{array}$$

will commute if $\alpha' = s \circ \alpha, \beta = \beta' \circ s$.

$\leadsto \theta_{(r_i)}^I: \mathcal{O}(\hat{G} \parallel (\hat{G} \rtimes Q)^I \parallel \hat{G}, \Delta) \rightarrow \text{End}(\text{id}_{\mathcal{D}(\text{Bun}_{\hat{G}})})$

$\leadsto \Theta_{n, (r_i)}: \mathcal{O}(\hat{G} \rtimes Q)^n \parallel \hat{G} \rightarrow \text{End}(\text{id}_{\mathcal{D}(\text{Bun}_{\hat{G}})})$
 $\quad \quad \quad \parallel$
 $\quad \quad \quad \mathbb{Z}^1(F_n, \hat{G}) \parallel \hat{G}$